



SOUTHEAST UNIVERSITY
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Project Report

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Project Name : HeritageLink: A Secure Web-Based Platform for Connecting Tourists with Tour Guides in Bangladesh

GitHub Repository Link : <https://github.com/SayedTasnimSinthia/HeritageLink-A-Secure-Web-Based-Platform-for-Connecting-Tourists-with-Tour-Guides.git>

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“HeritageLink: A Secure Web-Based Platform for Connecting Tourists with Tour Guides in Bangladesh”

Chapter 1 : Objectives & Scope

Project Overview :

Bangladesh possesses numerous archaeological sites, historical monuments & cultural destinations that draw domestic & international tourists. However, international visitors often struggle to locate local guides who possess targeted heritage site expertise & specific language interpretation skills. HeritageLink is a centralized, web-based platform that connects tourists directly with dual-role guides who serve concurrently as site specialists & translators. The platform features guide pricing, slot-based availability tracking, booking management & a dedicated administrative portal built with HTML5, CSS3, JavaScript, PHP & MySQL.

Objectives :

- Develop a centralized, web-based platform for heritage tour booking.
- Implement secure user authentication with role-based access control.
- Enable tourists to conveniently browse & book guided tours online.
- Maintain a registry of tour guides with dual-role (guide & translator) proficiencies.
- Enable tourists to search, filter & book guides based on target languages & heritage site expertise.
- Enable admin to manage site listings, oversee guide data & process tour bookings.

Project Scope :

Portal	Target Users	Core Scope & Capabilities
Tourist Portal	Registered Tourists	Registration, secure login, profile management, destination directory, guide search & filtering, slot booking, booking history & post-tour reviews.
Admin Portal	System Administrators	Guide verification & CRUD operations, destination catalog management, schedule management,

		customer auditing, booking status processing & subscriber records.
Public Pages	General Visitors	Landing page, destination directory, guide directory, About Us, FAQs, platform policies (Terms, Privacy, Guide Policy) & newsletter subscriptions.

Technology Stack :

Layer	Technology Used
Frontend	HTML5, CSS3, JavaScript
Backend	PHP
Database	MySQL
Web Server	Apache
Database Management	phpMyAdmin
Version Control	Git & GitHub
Development Environment	Visual Studio Code

Chapter 2 : Analysis Phase

The analysis phase identified user roles, core functional requirements, permission boundaries, data entities & security constraints prior to system implementation.

User Requirements :

Admin :

- Manage tour guide records.
- Manage heritage destination records.
- Manage guide availability schedules.
- View registered customers.
- Manage customer bookings.
- View newsletter subscribers.

Tourist :

- Register & log in.
- Manage personal profile information.
- Browse heritage destinations.
- Search & filter tour guides.
- View guide languages, specializations, locations & pricing.
- View guide availability.
- Book available guide schedules.
- View booking status & history.
- Submit a rating & review after a completed tour.

Functional Requirements :

- FR1: Provide secure authentication, role segregation & session management.
- FR2: Allow administrators to perform CRUD operations on guide profiles & license records.
- FR3: Allow administrators to manage heritage destination details, fees & operational statuses.
- FR4: Enable dynamic management & status tracking of guide availability slots.
- FR5: Allow tourists to browse all destination records.
- FR6: Enable tourists to search & filter guides by language, heritage site, district & rate.

- FR7: Display availability schedules to tourists.
- FR8: Allow tourists to book available schedules & prevent double-booking.
- FR9: Allow tourists to track booking status & historical records.
- FR10: Restrict rating (1-5) & review submissions exclusively to completed bookings.
- FR11: Provide admin with centralized tools to manage customer records & process bookings.
- FR12: Store visitor newsletter subscriptions.

Non-Functional Requirements :

The system ensures baseline security through hashed passwords at registration, session-based access control for tourist & admin areas & input validation on most forms, with parameterized queries used in the newer modules (guide registration, availability booking & reviews). It provides high usability & responsiveness through clean, mobile-friendly navigation across modern browsers (Chrome, Firefox, Safari, Edge). Performance & reliability are supported through optimized database queries, structured error handling & modular code across PHP, HTML, CSS & JavaScript.

Software Development Life Cycle (SDLC) Model Selection :

The Agile model was selected because the system has several interconnected modules that can be built, tested & improved incrementally.

Chapter 3 : Design & Implementation Phase

System Architecture :

HeritageLink follows a three-tier client-server architecture.

1. Presentation tier: HTML5, CSS3 & JavaScript provide the interface, forms, filtering & interactivity across both the tourist & admin portal.
2. Application tier: PHP handles authentication, sessions, business logic, CRUD operations, bookings & database communication.
3. Data tier: MySQL stores customer, guide, availability, destination, booking, review & subscriber data.

System Flow & Development Workflow :

The user submits a request through the browser, PHP processes the business logic & queries MySQL & the resulting data is rendered back into HTML for the browser to display. The development followed the same order; first the database was designed, then the frontend was built with HTML, CSS & JavaScript for pages, forms & filtering & finally PHP was added to handle login, CRUD operations & bookings, ending with testing across both the tourist & admin sides..

Authentication Design :

Tourists: stored in the customers table. Passwords are stored as hashes & verified at login & a session is created on success.

Administrator: uses a single predefined admin account verified in PHP & maintained through a session. Admin credentials are not stored in the database.

Database Schema :

1. subscribers (Tracks email newsletter sign-ups)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, email VARCHAR(255) NOT NULL UNIQUE, subscribed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

2. customers (Stores tourist accounts, credentials & emergency details)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) DEFAULT NULL, email VARCHAR(255) NOT NULL UNIQUE, password VARCHAR(255) NOT NULL, phone

VARCHAR(30) DEFAULT NULL, nid_number VARCHAR(50) DEFAULT NULL, profile_photo VARCHAR(255) DEFAULT 'images/default-avatar.png', gender ENUM('Male', 'Female', 'Other') DEFAULT NULL, dob DATE DEFAULT NULL, blood_group VARCHAR(10) DEFAULT NULL, nationality VARCHAR(100) DEFAULT 'Bangladeshi', emergency_name VARCHAR(150) DEFAULT NULL, emergency_phone VARCHAR(30) DEFAULT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

3. guides (Maintains multilingual tour guide informations)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, guide_id VARCHAR(50) NOT NULL UNIQUE, full_name VARCHAR(150) NOT NULL, email VARCHAR(150) NOT NULL UNIQUE, phone VARCHAR(30) NOT NULL, emergency_phone VARCHAR(30) DEFAULT NULL, gender ENUM('Male', 'Female', 'Other') NOT NULL, dob DATE NOT NULL, nid_number VARCHAR(50) NOT NULL UNIQUE, license_no VARCHAR(100) DEFAULT NULL, experience_years INT NOT NULL DEFAULT 0, specialization VARCHAR(255) NOT NULL, languages TEXT NOT NULL, heritage_sites TEXT NOT NULL, working_division VARCHAR(50) NOT NULL DEFAULT 'Dhaka', rate_type ENUM('Hourly', 'Daily') NOT NULL DEFAULT 'Daily', rate_amount DECIMAL(10,2) NOT NULL DEFAULT 0.00, profile_photo VARCHAR(255) DEFAULT 'images/default-avatar.png', short_bio TEXT DEFAULT NULL, account_status ENUM('Active', 'Inactive') NOT NULL DEFAULT 'Active', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

4. guide_availability (Tracks guide scheduling slots, locations & capacity)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, availability_id VARCHAR(50) NOT NULL UNIQUE, guide_id VARCHAR(50) NOT NULL, available_date DATE NOT NULL, start_time TIME NOT NULL, end_time TIME NOT NULL, working_district VARCHAR(100) NOT NULL, working_sites TEXT NOT NULL, max_tourists INT NOT NULL DEFAULT 10, slot_status ENUM('Available', 'Booked', 'On Hold', 'Unavailable') NOT NULL DEFAULT 'Available', notes TEXT DEFAULT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, CONSTRAINT fk_guide_avail_guide_id FOREIGN KEY (guide_id) REFERENCES guides(guide_id) ON DELETE CASCADE ON UPDATE CASCADE

5. destinations (Stores cataloged cultural & heritage destinations)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, destination_id VARCHAR(50) NOT NULL UNIQUE, site_name VARCHAR(150) NOT NULL, division VARCHAR(50) NOT NULL, district VARCHAR(100) NOT NULL, location VARCHAR(255) NOT NULL, heritage_type VARCHAR(150) NOT NULL, historical_period VARCHAR(100) NOT NULL, built_year VARCHAR(50) DEFAULT NULL, opening_hours VARCHAR(100) NOT NULL, entry_fee VARCHAR(255) NOT NULL DEFAULT 'Free', guide_available ENUM('Yes', 'No') NOT NULL

DEFAULT 'No', destination_image VARCHAR(255) DEFAULT 'images/default-destination.jpg', description TEXT NOT NULL, status ENUM('Active', 'Under Maintenance', 'Closed') NOT NULL DEFAULT 'Active', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

6. bookings (Stores reservation transactions linking customers, guides & specific slots.)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, booking_id VARCHAR(50) NOT NULL UNIQUE, customer_id INT NOT NULL, guide_id VARCHAR(50) NOT NULL, availability_id VARCHAR(50) NOT NULL, site_name VARCHAR(150) NOT NULL, booking_date DATE NOT NULL, time_slot VARCHAR(50) NOT NULL, status ENUM('Pending', 'Confirmed', 'Cancelled', 'Completed') NOT NULL DEFAULT 'Pending', created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, CONSTRAINT fk_booking_customer FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE, CONSTRAINT fk_booking_guide FOREIGN KEY (guide_id) REFERENCES guides(guide_id) ON DELETE CASCADE

7. reviews (Stores tourist feedback & ratings for completed tours)

Attributes: id INT AUTO_INCREMENT PRIMARY KEY, booking_id VARCHAR(50) NOT NULL UNIQUE, customer_id INT NOT NULL, guide_id VARCHAR(50) NOT NULL, rating INT NOT NULL CHECK (rating >= 1 AND rating <= 5), review_text TEXT DEFAULT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP, CONSTRAINT fk_review_booking FOREIGN KEY (booking_id) REFERENCES bookings(booking_id) ON DELETE CASCADE, CONSTRAINT fk_review_customer FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE, CONSTRAINT fk_review_guide FOREIGN KEY (guide_id) REFERENCES guides(guide_id) ON DELETE CASCADE

Database Relationships

The HeritageLink database maintains referential integrity through foreign-key constraints.

- One customer can create multiple bookings.
- One guide can have multiple guide availability records.
- One guide can receive multiple bookings.
- One booking belongs to one customer & one guide.
- One booking can have only one review.
- One review belongs to one customer & one guide.
- Newsletter subscribers are stored independently in the subscribers table.

Chapter 4 : Testing Phase

The system underwent black-box functional testing to validate input handling, state transitions, authentication barriers & database transactions.

Test ID	Module	Test Condition	Expected Behavior	Result
TC01	Auth	Register valid tourist credentials	Record inserted into customers; password hashed	Pass
TC02	Auth	Register duplicate email address	Constraint failure intercepted; user-friendly error shown	Pass
TC03	Auth	Valid login submission	Session established; redirect to dashboard	Pass
TC04	Auth	Invalid password attempt	Authentication denied; session not created	Pass
TC05	Profile	Update profile	customers table updated successfully	Pass
TC06	Catalog	Browse destination directory	Display all records	Pass
TC07	Discovery	Filter guides by language & location	Returns guides matching target query criteria	Pass
TC08	Booking	Reserve an Available guide slot	Record added to bookings as Pending; slot updated to On Hold	Pass

TC09	History	Open tourist booking history panel	Displays linked records matching customer_id	Pass
TC10	Review	Submit review for completed booking	Review recorded with rating (1-5)	Pass
TC11	Review	Attempt duplicate review on same booking	Blocked by unique key constraint on booking_id	Pass
TC12	Review	Submit rating outside range (<1 or >5)	Blocked by schema CHECK constraint	Pass
TC13	Admin	Create guide profile with all fields	Record stored in guides with Active status	Pass
TC14	Admin	Update booking status to Completed	bookings.status changed; review eligibility unlocked	Pass
TC15	Newsletter	Submit duplicate subscription email	Handled gracefully without DB crash due to unique key	Pass

Chapter 5 : Reflection, Future Enhancement & Limitations

Reflection & Future Enhancements :

Building the system showed that a well designed database keeps data safe & stops many errors before they happen & checking user inputs helps reduce bad data & basic hacking attempts. To reduce double booking, the system marks a guide's time slot on hold as soon as a tour is reserved & unique booking rules make sure tourists can only leave one review per completed trip. Looking ahead, future plans include adding easy online payments through bKash & bank cards, direct in-app messaging between tourists & their guides, interactive map directions, dedicated mobile apps for Android & iOS & automated tools to verify official guide licenses quickly.

Limitations :

- Login authentication has a legacy fallback path that is less secure than the primary hashing mechanism.
- The platform lacks CSRF protection on forms that create, update or delete data.
- Guide availability status is refreshed periodically rather than updated instantly as bookings change.
- Guide verification is based on manually entered records rather than an automated credential checking process.
- Deleting a guide account also removes their related booking & review history, rather than archiving it.

Chapter 6 : References

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